

```
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving SampleSuperstore.csv to SampleSuperstore.csv

```
import pandas as pd

df = pd.read_csv('SampleSuperstore.csv')

df.head()
```

	Ship Mode	Segment	Country	City	State	Postal Code	Region	Category	Sub-Category	Sales
0	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Bookcases	261.90
1	Second Class	Consumer	United States	Henderson	Kentucky	42420	South	Furniture	Chairs	731.90
2	Second Class	Corporate	United States	Los Angeles	California	90036	West	Office Supplies	Labels	14.60
3	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Furniture	Tables	957.50
4	Standard Class	Consumer	United States	Fort Lauderdale	Florida	33311	South	Office Supplies	Storage	22.30

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Ship Mode             9994 non-null   object  
1   Segment               9994 non-null   object  
2   Country               9994 non-null   object  
3   City                 9994 non-null   object  
4   State                9994 non-null   object  
5   Postal Code          9994 non-null   int64   
6   Region               9994 non-null   object  
7   Category             9994 non-null   object  
8   Sub-Category         9994 non-null   object  
9   Sales                9994 non-null   float64  
10  Quantity             9994 non-null   int64   
11  Discount             9994 non-null   float64  
12  Profit              9994 non-null   float64  
dtypes: float64(3), int64(2), object(8)
memory usage: 1015.1+ KB
```

```
df.describe()
```

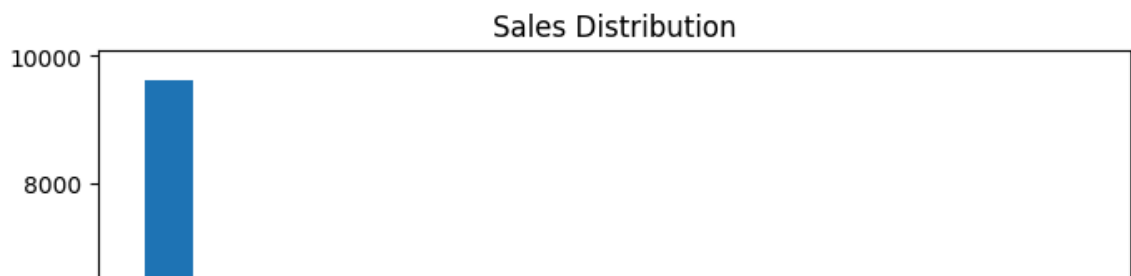
	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	55190.379428	229.858001	3.789574	0.156203	28.656896
std	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	90008.000000	209.940000	5.000000	0.200000	29.364000
max	99301.000000	22638.480000	14.000000	0.800000	8399.976000

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.hist(df['Sales'], bins=20)

plt.title('Sales Distribution')
plt.xlabel('Sales')
plt.ylabel('Frequency')

plt.show()
```



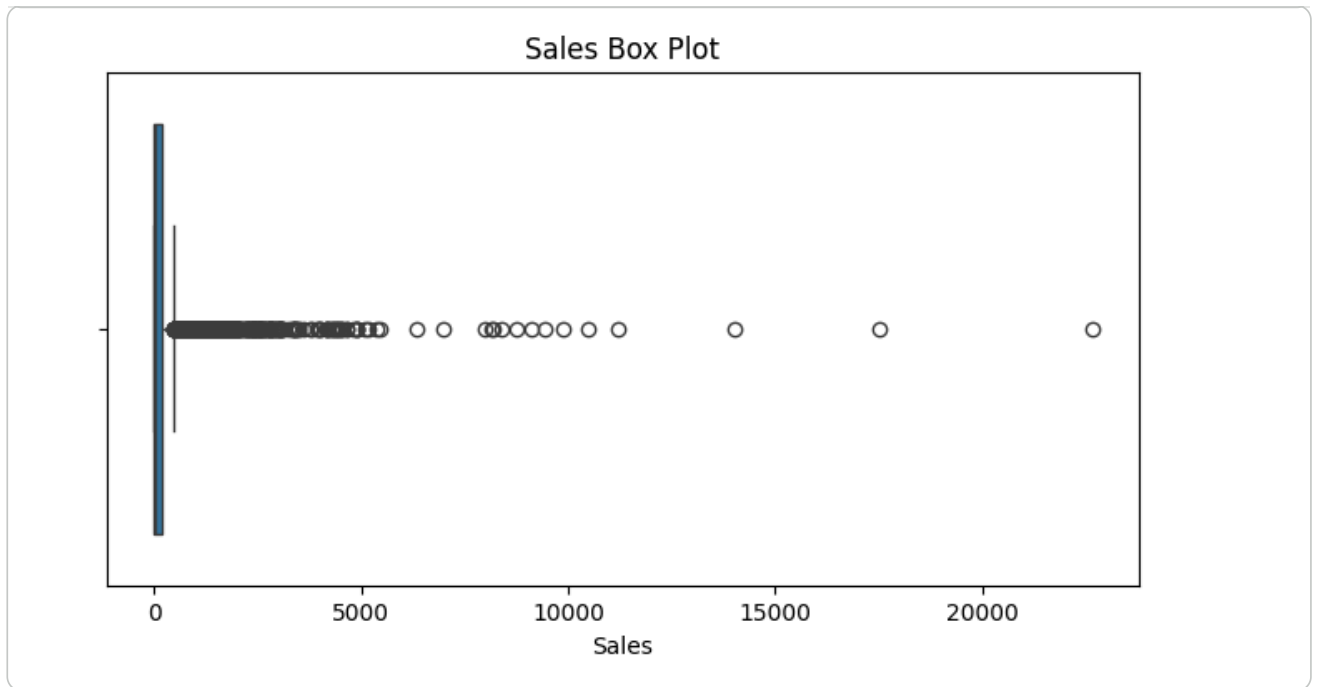
```
import seaborn as sns

plt.figure(figsize=(8,4))

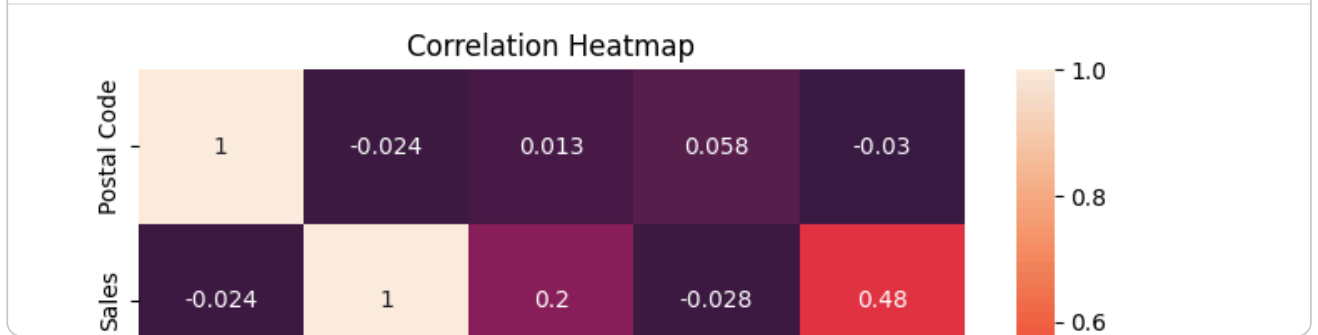
sns.boxplot(x=df['Sales'])

plt.title('Sales Box Plot')

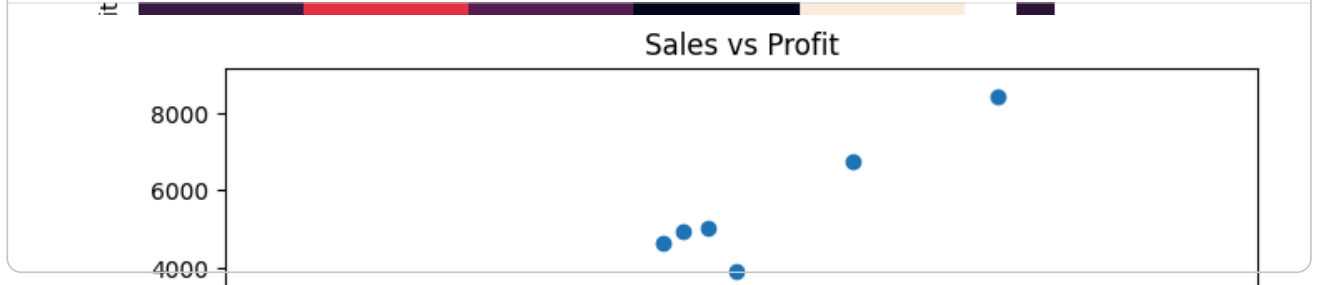
plt.show()
```



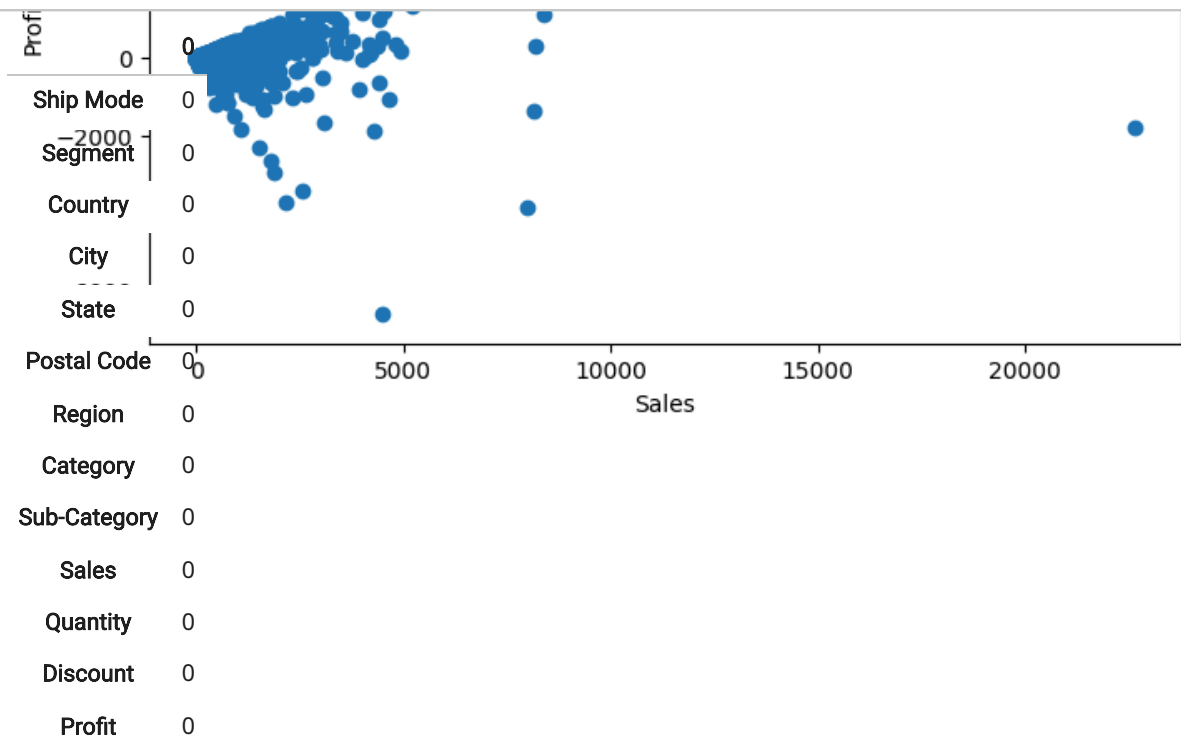
```
plt.figure(figsize=(8,6))  
  
sns.heatmap(df.corr(numeric_only=True), annot=True)  
  
plt.title('Correlation Heatmap')  
  
plt.show()
```



```
plt.figure(figsize=(8,5))  
  
plt.scatter(df['Sales'], df['Profit'])  
  
plt.xlabel('Sales')  
plt.ylabel('Profit')  
  
plt.title('Sales vs Profit')  
  
plt.show()
```



```
df.isnull().sum()
```



dtype: int64

```
df.drop_duplicates(inplace=True)
print("Duplicates removed")
```

Duplicates removed

```
df.to_csv('cleaned_superstore.csv', index=False)
print("Cleaned dataset saved")
```

Cleaned dataset saved

```
from google.colab import files
files.download('cleaned_superstore.csv')
```